

EXERCISE 01

SQL FOUNDATIONS

SQL Fundamentals : SELECT & Filtering

①

Answer: `SELECT *`
`FROM employees;`

Output:

TABLE employees

id	First_name	Last_name	department	Salary	Hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-01-18	Durban
4	David	Mokoena	Marketing	80000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

② Answer: `SELECT DISTINCT department`
`FROM employees;`

Output:

Department
IT
HR
Finance
Marketing

③ Answer: `SELECT first_name, last_name`
`FROM employees`
`ORDER BY salary DESC;`
↓

Output:

First name	Last name
Eva	Waidoo
Alice	Green
Cathy	Zulu
David	Mokene
Brian	Lee

Q4 Answer: SELECT id, First name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

Output

id	First name	Last name	Salary
5	Eva	Waidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Q5 Answer: SELECT id, First name, Last name, department
FROM employees
WHERE department = 'IT';

Output

id	First name	Last name	Department
1	Alice	Green	IT
5	Eva	Waidoo	IT

Q6 Answer: SELECT id, First name, Last name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 60000;

Output:

id	First_name	Last_name	Department	Salary
3	Cathy	Turner	Finance	65000

⑦ Answer: SELECT id, First_name, Last_name, Department
FROM employees
WHERE Department IN ('HR', 'Marketing');

Output

id	First_name	Last_name	Department
2	Brian	Lee	HR
4	David	Mukherjee	Marketing

⑧ Answer: SELECT id, First_name, Last_name, Department
FROM employees
WHERE Department <> 'IT';

Output

id	First_name	Last_name	Department
2	Brian	Lee	HR
3	Cathy	Turner	Finance
4	David	Mukherjee	Marketing

⑨ SELECT id, First_name, Last_name, Department
FROM employees
WHERE Department IN ('IT', 'HR', 'Finance');

Output

id	First_name	Last_name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Turner	Finance
5	Eva	Murphy	IT

⑩ Answer: SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 68000
AND city = 'Johannesburg';

output

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Meißen	IT	72000	Johannesburg